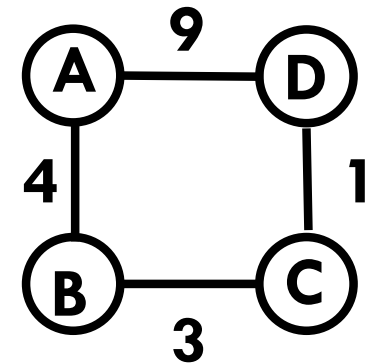


Quiz

16

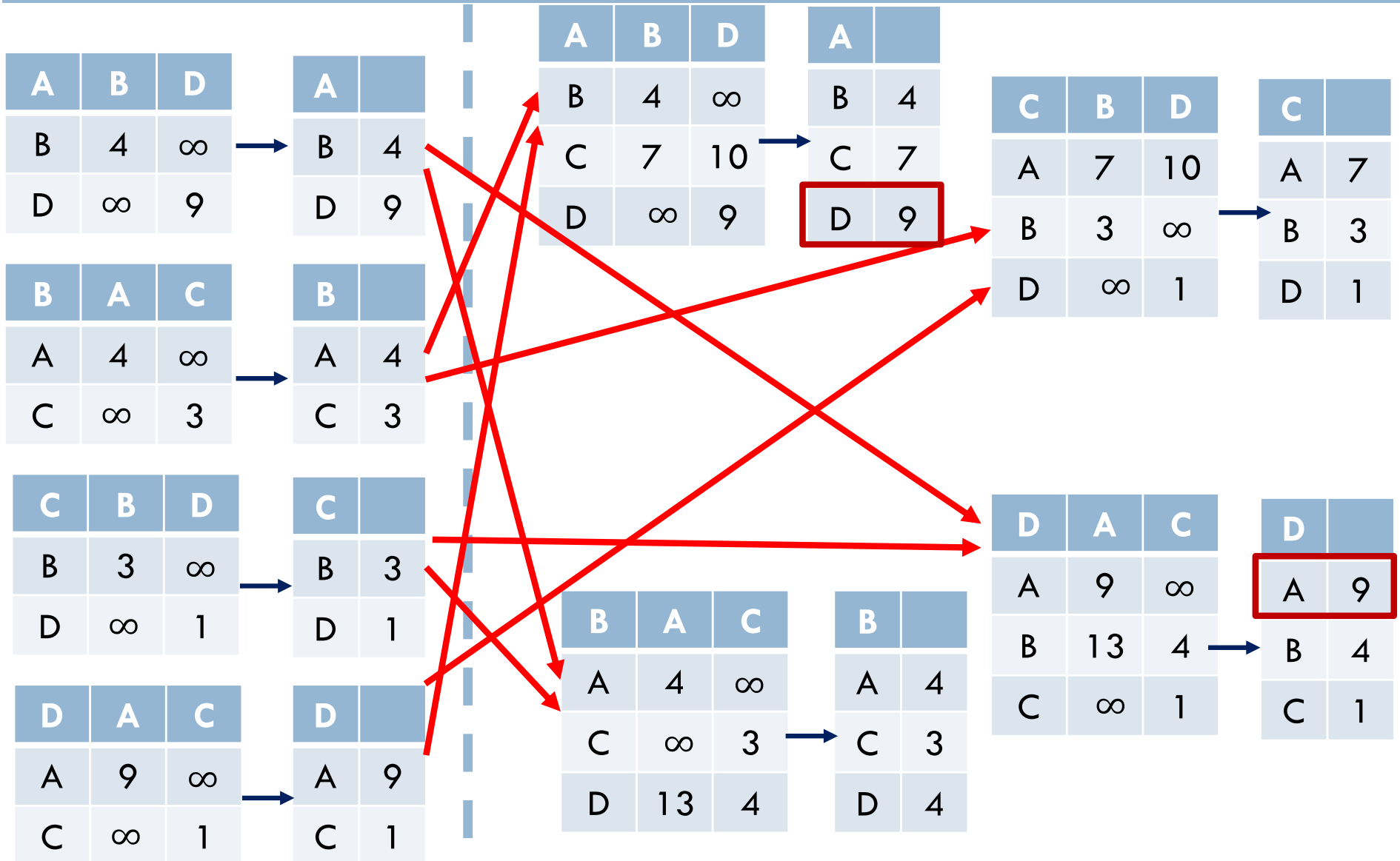
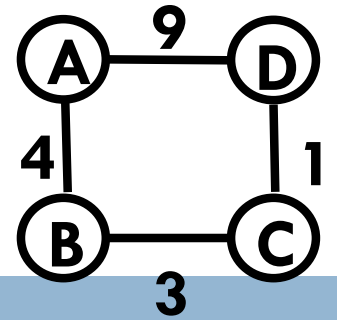
- Consider a network topology with three nodes shown in the right. All nodes are running the distance vector routing algorithm to find the shortest path to other nodes.

How many rounds does it take for the algorithm to converge? Feel free to draw the distance vector tables at each round.



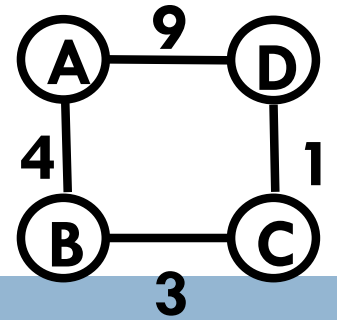
Quiz

17



Quiz

18

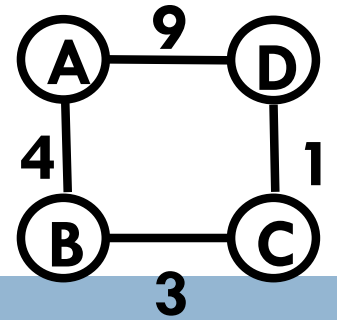


A	B	D		
B	4	13	B	4
C	7	10	C	7
D	8	9	D	8

D	A	C		
A	9	8	A	8
B	13	4	B	4
C	16	1	C	1

Quiz

19



- A quicker method:
 - ▣ Find the shortest path for each pair of node
 - ▣ Among all the shortest paths, identify the one with the most hops
 - ▣ The highest hop count minus 1 is the # of rounds taken to converge

- This assumes:
 - ▣ Synchronous rounds
 - ▣ No count-to-infinity
 - ▣ Static topology and no triggered update

Data Link Layer Services

20

- **Framing**

- ▣ Add header and trailer, encapsulate packet into frame

- **Error detection and correction**

- ▣ Errors caused by signal attenuation, noise

- **Medium access control**

- ▣ Who can access the medium and when

- **Reliable delivery**

- ▣ Local retransmission upon frame losses

Two Types of Link Media

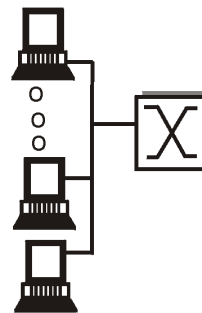
21

□ Point-to-point

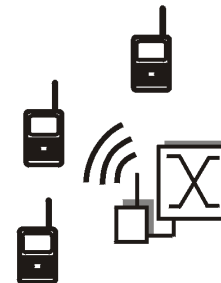
- Long-distance fiber link
- Point-to-point link between Ethernet switch and host

□ Broadcast (shared wire or medium)

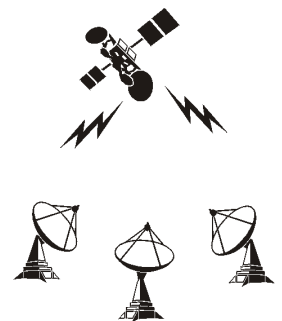
- Traditional Ethernet
- 802.11 wireless LAN



shared wire
(e.g. Ethernet)



shared wireless
(e.g. Wavelan)



satellite

Share a Medium

22

- Avoid multiple nodes speaking at once
- **Medium Access Control (MAC) protocol**
 - ▣ Distributed algorithm for sharing the medium
 - ▣ Determine whether the node can transmit
- What're the possible ways of sharing?

Blah, blah, blah



ZZZZZZZZZZZZZZZZZZ



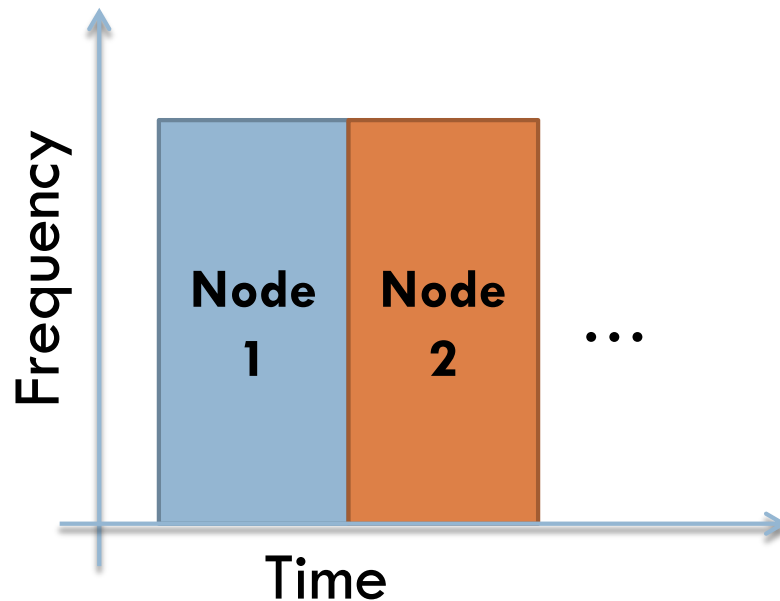
cocktail party

#1: Channel Partitioning

23

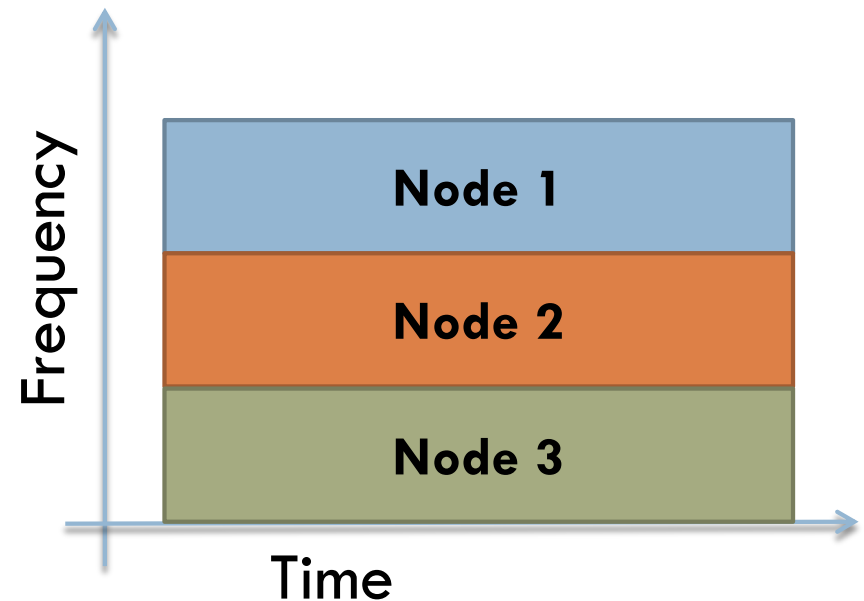
Time Division Multiple Access (TDMA)

- Each node gets a fixed-length time slot in each round



Frequency Division Multiple Access (FDMA)

- Each node gets a fixed frequency band in all time

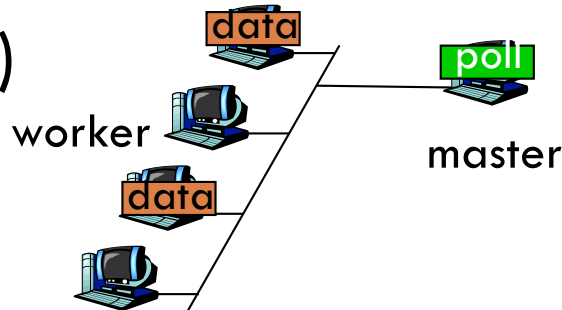


#2: “Taking Turns”

24

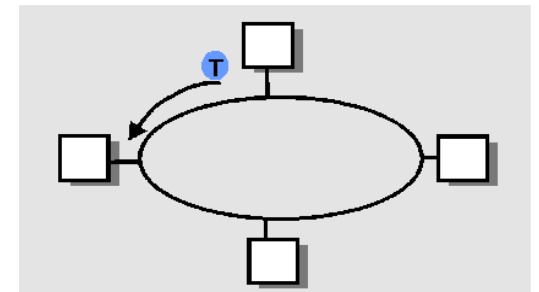
Polling

- Master node invites workers to transmit
- Cons
 - ▣ Polling overhead
 - ▣ Latency
 - ▣ Single point of failure (master)



Token Passing

- Pass a control token from one node to next
- Cons
 - ▣ Token overhead
 - ▣ Latency
 - ▣ Single point of failure (token)



How Do You Like These So far?

25

- Controlled access can result into channel waste
 - ▣ Especially given the bursty, dynamic traffic
- Any idea to do better?
- Do not avoid collisions, just recover from them!

#3: Random Access Protocol

26

- No fixed schedule, no special node to coordinate
- When a node has packets to send, transmits at the full channel rate
- Collision occurs when two or more transmit at the same time
- Key components
 - ▣ How to detect collisions?
 - ▣ How to recover from collisions?

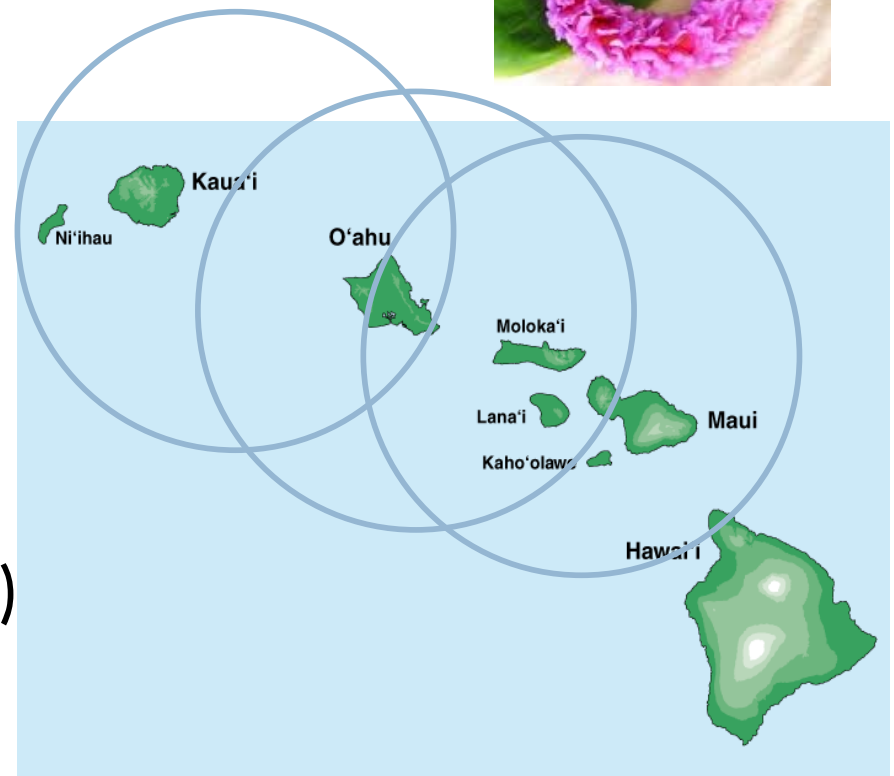
Start with the Simplest

27

□ Slotted ALOHA

- In Hawaii language, ALOHA means affection, love, peace, compassion, goodbye, and hello!

- Invented by Prof. Norm Abramson (1970)
 - Set up the first radio communication system for Hawaiian islands! (AlohaNet)



Slotted ALOHA

28

□ Assumptions

- ▣ Divide time into equal slots, one frame per slot
- ▣ Nodes start to transmit only at the start of slots, nodes are synchronized
- ▣ Can detect collision right away

□ Operations

- ▣ Have data? Just send
- ▣ No collision: send a new frame in the next slot
- ▣ Collision: retransmit frame in the next slot with probability p

Slotted ALOHA Example

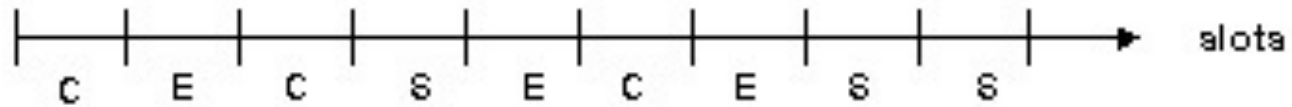
29



node 1

node 2

node 3



Efficiency of Slotted ALOHA

30

- Suppose N nodes have packets to send
 - ▣ Each transmits in a slot with probability p
- Probability of successful transmission:
 - by a **particular** node i : $S_i = p (1-p)^{(N-1)}$
 - by **any** of N nodes: $S = N p (1-p)^{(N-1)}$
- What value of p maximizes the prob. of success:
 - ▣ For fixed p , $S \rightarrow 0$ as N increases
 - ▣ But if **$p = 1/N$** , then $S \rightarrow 1/e = 0.37$ as N increases
- **Max efficiency is only slightly greater than $1/3$!**

Slotted ALOHA: The Good and Bad

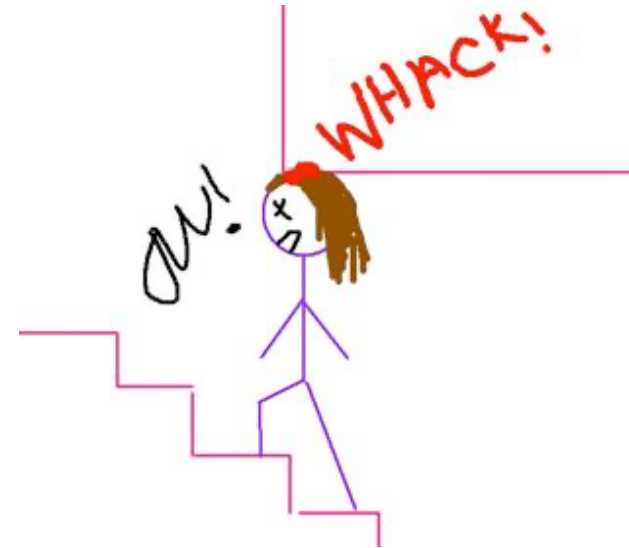
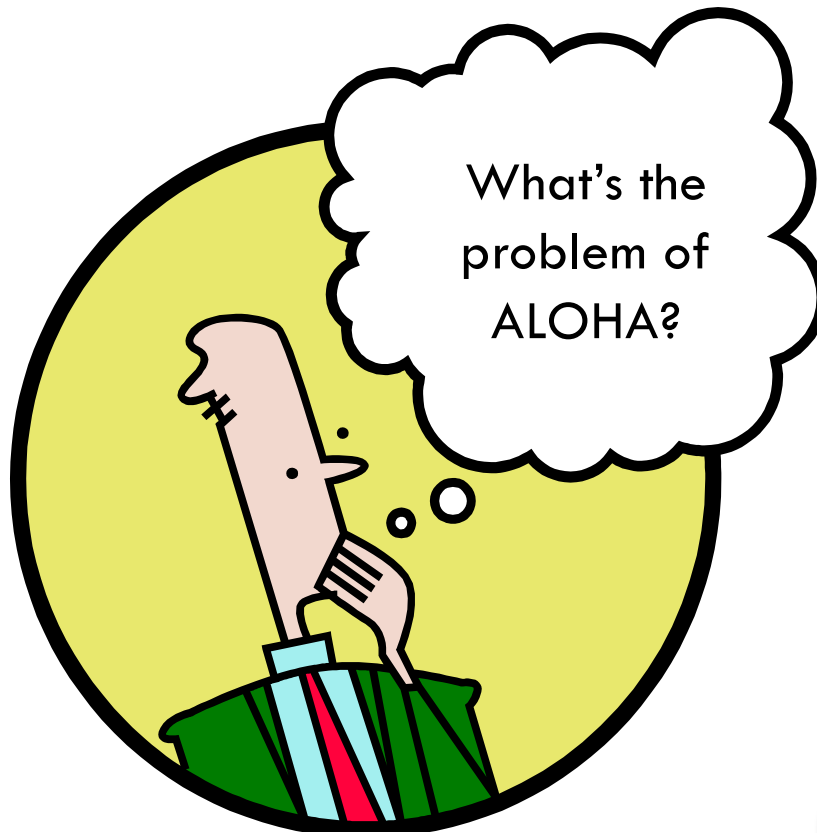
31

□ Pros

- Simple
- Single active node can continuously transmit at full rate
- Highly decentralized

□ Cons

- Collisions
- Idle slots
- Synchronization



Doesn't sense the medium,
just send send send...

Listen before Transmit

33

- **Carrier Sense Multiple Access (CSMA)**

- If channel sensed idle, transmit entire frame
- If channel sensed busy, defer transmission



- Problem solved?

- No... collision can still occur

- Propagation delay: nodes do not hear each other immediately
- Or the transmitter cannot hear at all!

CSMA/CD (Collision Detection)

34

- Carrier sensing, as in CSMA
 - ▣ **Collisions detected within short time**
 - ▣ Colliding transmissions aborted, reducing wastage
- Collision detection easy in wired (broadcast) LANs
 - ▣ Compare transmitted, received signals
- Collision detection difficult in wireless LANs

Collision Detection Example

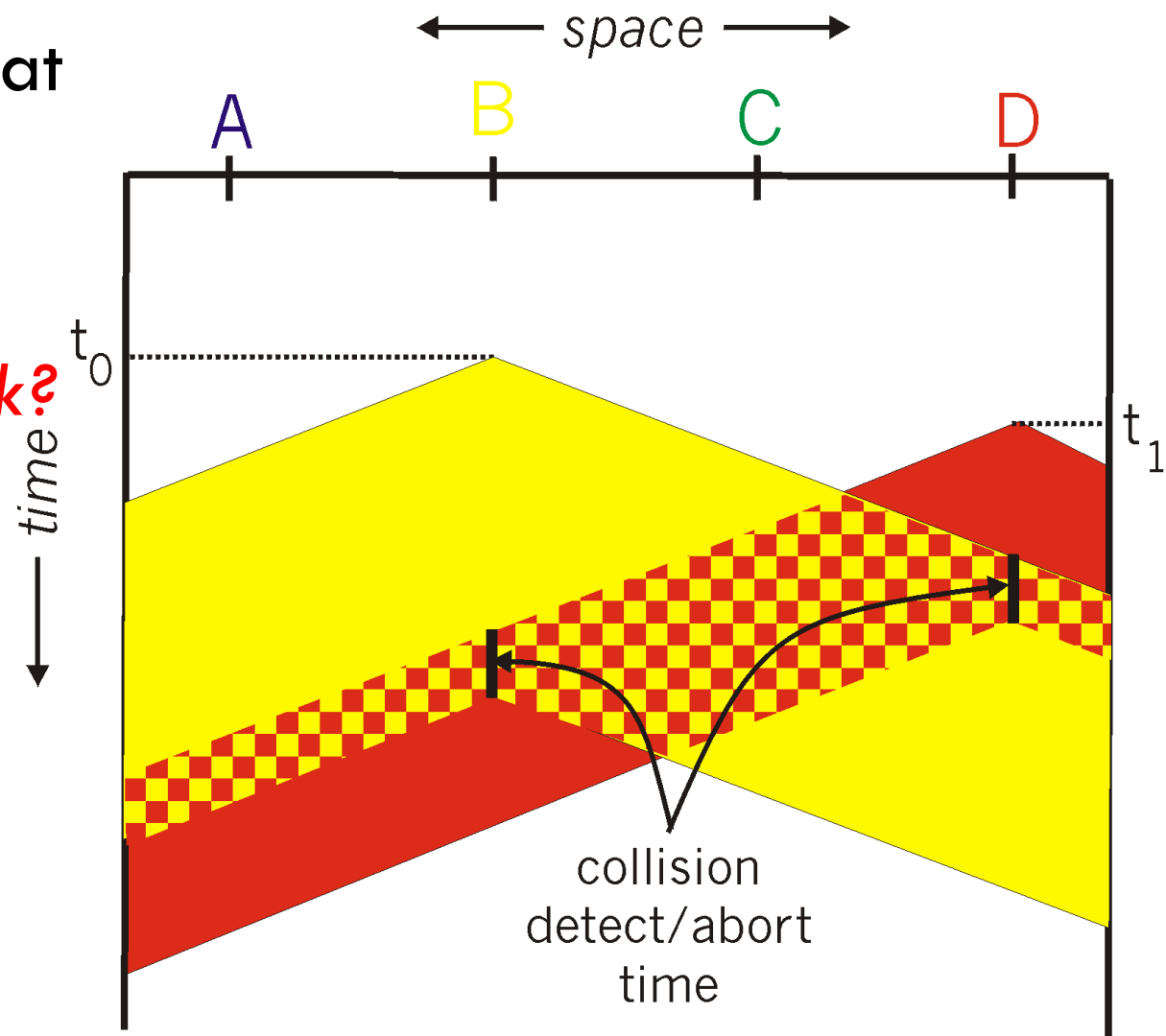
35

□ B and D can tell that collision occurred

□ *Does it always work?*

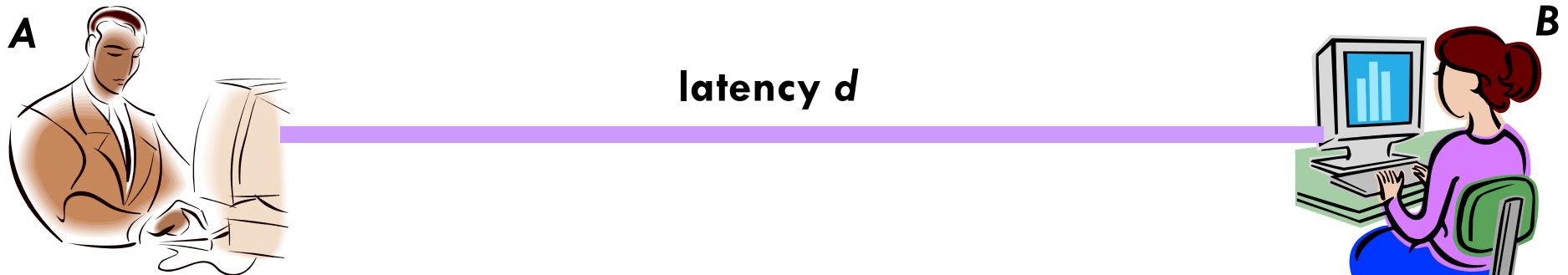
▣ *Packet cannot be too small*

▣ *Distance cannot be too long*



Limits on CSMA/CD Network Length

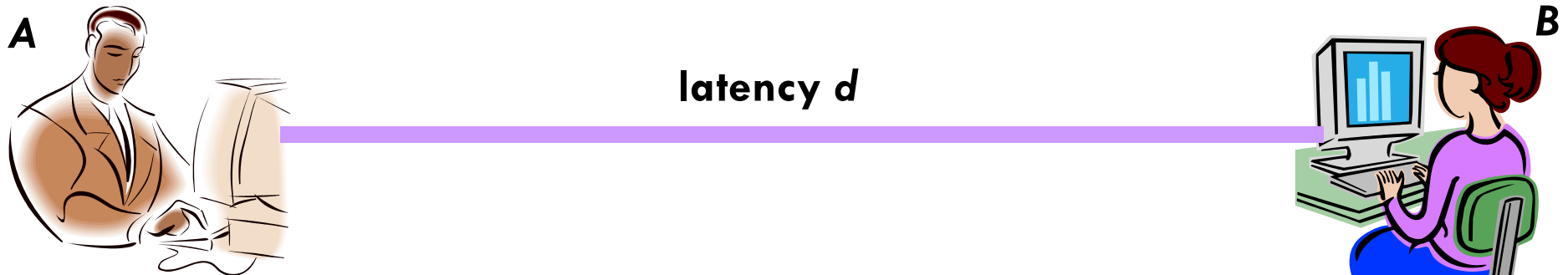
36



- Latency depends on physical length of link
 - ▣ Time to propagate a packet from one end to the other
- A sends a packet at time t
 - ▣ B sees an idle line until $t+d$
- B sends a packet just before A's signal arrives
- Signals overlap → Collision
- Distorted waveform propagate back after another d
- The longest delay for A to detect a collision: $2d$

Limits on CSMA/CD Network Length

37



- A needs to wait for time $2d$ to detect collision!
 - ▣ IEEE 802.3 Ethernet specifies max $2d$ to be $51.2\mu\text{s}$
- Impose restrictions, e.g., 10Mbps Ethernet
 - ▣ Min length of the frame: 512 bits
 - ▣ Max length of Ethernet copper cable: 100m (mostly due to signal attenuation)

Key Ideas of Random Access

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□ **Carrier sense**

- *Listen before speaking, and do not interrupt*
- Checking if someone else is already sending data, wait till the other node is done

□ **Collision detection**

- Realizing when two nodes are transmitting at once by detecting that the data on the wire is garbled

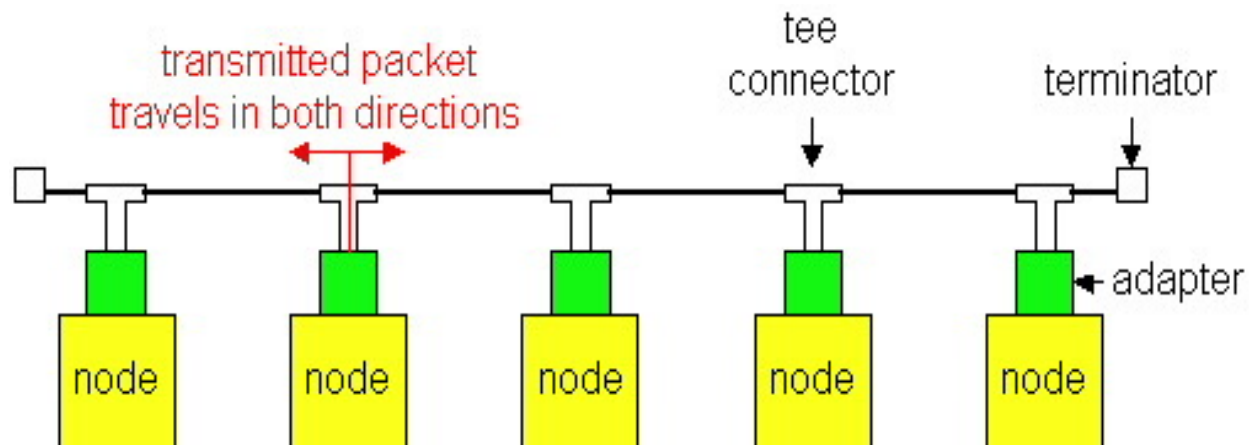
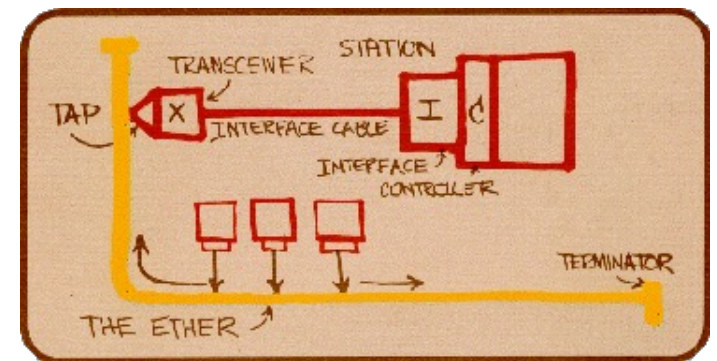
□ **Randomness (Collision recovery)**

- *Do not start talking again right away*
- Waiting for a random time before trying again

Ethernet

39

- Started with a shared wired medium

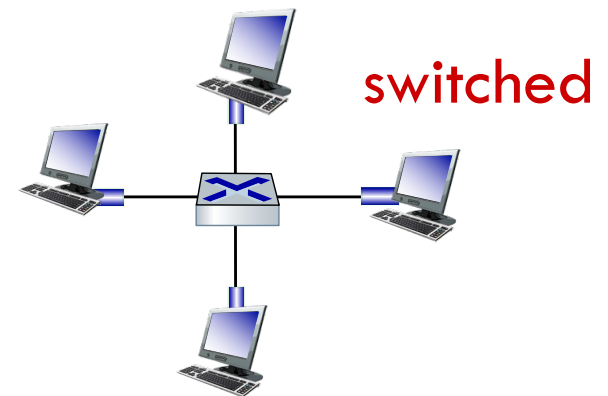


Ethernet Evolution

40

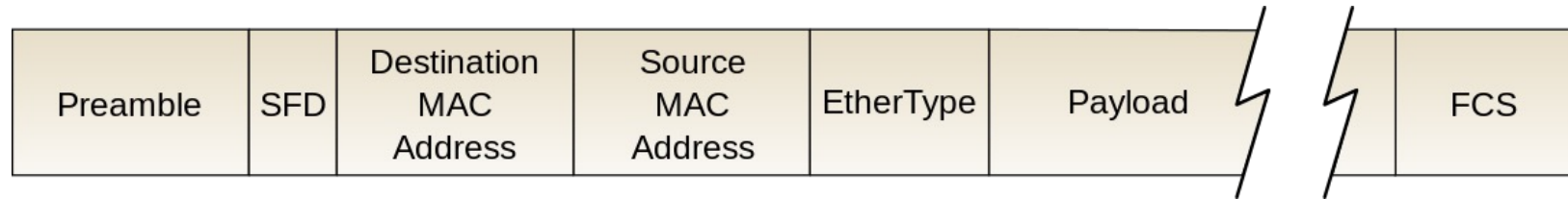
- Ethernet was invented as a broadcast technology
 - ▣ Hosts share channel
 - ▣ Each packet received by all attached hosts
 - ▣ Unslotted CSMA/CD with binary backoff for media access control

- Current Ethernets are “switched”
 - ▣ Point-to-point **full-duplex** links between switches, between a host and switch
 - ▣ No sharing, no CSMA/CD



Ethernet Frame Structure

41



- ❑ **Preamble:** 7 bytes with a particular pattern used to synchronize receiver, sender clock rates
- ❑ **Addresses:** 6 bytes, frame is received by all adapters on a LAN and dropped if address does not match
- ❑ **Type:** 2 bytes, indicating higher-layer protocol (e.g., IP, Appletalk)
- ❑ **CRC:** 4 bytes for error detection
- ❑ **Data payload:** 46 – 1500 Bytes

Medium Access Control (MAC) Address

42

- 6 bytes, e.g., 00-15-C5-49-04-A9 in hexadecimal
- Unique, hard-coded in the adapter when it is built
- Hierarchical allocation
 - ▣ **Blocks**: assigned to vendors (e.g., Dell) by the IEEE
 - First 24 bits (e.g., 00-15-C5-**-**-**)
 - ▣ **Adapter**: assigned by the vendor from its block
 - Last 24 bits
- Broadcast address: FF-FF-FF-FF-FF-FF
- **Q: What're the differences between MAC address and IP address?**

Two Types of MAC Address

43

- Burned-in MAC address
 - ▣ Stored in hardware and assigned by manufacturer
 - ▣ Globally unique in principle
- Effective MAC address used on the network
 - ▣ OS can configure
 - ▣ Can be overridden (MAC spoofing)
 - ▣ Used for privacy randomization, virtualization, etc.
- Network does not rely on MAC addresses for security

Bootstrap Communication

44

- DHCP
- ARP

Quiz

45

- Let's define bandwidth-latency product (BDP) as the multiplication of the link bandwidth and the round-trip time delay. For the midterm network on the right, what is the BDP? Speed of sound is 1.5km/s , and speed of light is $3 \times 10^8\text{m/s}$.

